

Task 30: Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit

Problem Description

Given an array of integers `nums` and an integer `limit`, return the size of the longest non-empty subarray such that the absolute difference between any two elements of this subarray is less than or equal to `limit`.

Java Solution

```
class Solution {
    public int longestSubarray(int[] nums, int limit) {
        Deque<Integer> maxD = new ArrayDeque<>();
        Deque<Integer> minD = new ArrayDeque<>();
        int left = 0, res = 0;

        for (int right = 0; right < nums.length; right++) {
            while (!maxD.isEmpty() && maxD.peekLast() < nums[right])
                maxD.pollLast();
            while (!minD.isEmpty() && minD.peekLast() > nums[right])
                minD.pollLast();

            maxD.offer(nums[right]);
            minD.offer(nums[right]);

            while (maxD.peek() - minD.peek() > limit) {
                if (maxD.peek() == nums[left]) maxD.poll();
                if (minD.peek() == nums[left]) minD.poll();
                left++;
            }

            res = Math.max(res, right - left + 1);
        }
        return res;
    }
}
```